

Question ID: 56e373b3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

A manager is responsible for ordering supplies for a shaved ice shop. The shop's inventory starts with **4,500** paper cups, and the manager estimates that **70** of these paper cups are used each day. Based on this estimate, in how many days will the supply of paper cups reach **1,700**?

Answer

- A. **20**
- B. **40**
- C. **60**
- D. **80**

Correct Answer: B

Rationale

Choice B is correct. It's given that the shop's inventory starts with **4,500** paper cups and that the manager estimates that **70** of these paper cups are used each day. Let x represent the number of days in which the estimated supply of paper cups will reach **1,700**. The equation $4,500 - 70x = 1,700$ represents this situation. Subtracting **4,500** from both sides of this equation yields $-70x = -2,800$. Dividing both sides of this equation by -70 yields $x = 40$. Therefore, based on this estimate, the supply of paper cups will reach **1,700** in **40** days.

Choice A is incorrect. After **20** days, the estimated supply of paper cups would be $4,500 - 70(20)$, or **3,100** cups, not **1,700** cups.

Choice C is incorrect. After **60** days, the estimated supply of paper cups would be $4,500 - 70(60)$, or **300** cups, not **1,700** cups.

Choice D is incorrect. After **80** days, the estimated supply of paper cups would be $4,500 - 70(80)$, or $-1,100$ cups, which isn't possible.